Inequalities

Problem 1. Let n be a positive integer. Suppose each of $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ is a permutation of $1, \frac{1}{2}, \cdots, \frac{1}{n}$ such that

$$a_1 + b_1 \geqslant a_2 + b_2 \geqslant \cdots \geqslant a_n + b_n.$$

Prove that $a_k + b_k \leqslant \frac{4}{k}$ for each $k = 1, 2, \ldots, n$.

Problem 2. Prove that for any positive integers $a_1, a_2, \ldots, a_n$ the following inequality holds true:

$$\left\lfloor \frac{a_1^2}{a_2} \right\rfloor + \left\lfloor \frac{a_2^2}{a_3} \right\rfloor + \cdots + \left\lfloor \frac{a_n^2}{a_1} \right\rfloor \geqslant a_1 + a_2 + \cdots + a_n.$$

Problem 3. Let $x_1, x_2, \ldots, x_n$ be real numbers satisfying $0 \leqslant x_j \leqslant 1$ for each j. Prove that

$$(1 - x_1 x_2 \cdots x_n)^m + \big(1 - (1 - x_1)^m\big)\big(1 - (1 - x_2)^m\big) \cdots \big(1 - (1 - x_n)^m\big) \geqslant 1$$

for any positive integer m.

Problem 4. Let $k > 1$ be a real number, $n \geqslant 3$ be an integer, and $x_1 \geqslant x_2 \geqslant \cdots \geqslant x_n$ be positive real numbers. Prove that

$$\frac{x_1 + kx_2}{x_2 + x_3} + \frac{x_2 + kx_3}{x_3 + x_4} + \cdots + \frac{x_n + kx_1}{x_1 + x_2} \geqslant \frac{n(k+1)}{2}.$$

Problem 5. Let $A_1, A_2, \ldots, A_m$ be m subsets of a set of size n. Prove that

$$\sum_{i=1}^{m} \sum_{j=1}^{m} |A_i| \cdot |A_i \cap A_j| \geqslant \frac{1}{mn} \left(\sum_{i=1}^{m} |A_i| \right)^3.$$